later_seen | value-weighted | Absolute loss surfaces
Expanding controls are deduplicated sidebands, not finite coordinates
Exploratory seen-period estimates; calibration/power deferred; no calibrated significance claim

Separate | A: matched PCA

PC count K	52	78	104	130	156	182	208	234	260	286	312	338	364	390	416	442	468	494	520
8	0.877	0.819	0.774	0.745	0.756	0.745	0.747	0.737	0.734	0.747	0.751	0.757	0.749	0.744	0.742	0.740	0.740	0.742	0.744
7	0.860	0.810	0.777	0.740	0.744	0.739	0.744	0.737	0.734	0.747	0.750	0.755	0.748	0.749	0.744	0.743	0.743	0.744	0.746
6	0.836	0.801	0.767	0.740	0.748	0.739	0.747	0.737	0.734	0.744	0.749	0.750	0.746	0.742	0.742	0.744	0.745	0.746	0.748
5	0.825	0.804	0.764	0.732	0.741	0.741	0.743	0.735	0.730	0.742	0.748	0.746	0.746	0.738	0.745	0.751	0.750	0.753	0.755
4	0.828	0.791	0.761	0.734	0.737	0.738	0.737	0.734	0.722	0.738	0.735	0.747	0.749	0.748	0.746	0.745	0.748	0.752	0.755
3	0.820	0.782	0.758	0.735	0.729	0.737	0.743	0.742	0.735	0.744	0.746	0.748	0.746	0.744	0.743	0.742	0.742	0.744	0.748
2	0.796	0.783	0.753	0.732	0.731	0.744	0.744	0.743	0.733	0.737	0.743	0.745	0.745	0.743	0.744	0.743	0.739	0.738	0.741
1	0.769	0.790	0.772	0.745	0.745	0.746	0.746	0.743	0.736	0.745	0.756	0.757	0.758	0.756	0.757	0.755	0.751	0.748	0.752

Coefficient-history pairs

Separate | B: expanding PCA

PC count K	52	78	104	130	156	182	208	234	260	286	312	338	364	390	416	442	468	494	520
8	0.928	0.839	0.798	0.777	0.778	0.769	0.762	0.757	0.749	0.764	0.764	0.762	0.757	0.757	0.753	0.750	0.748	0.747	0.749
7	0.906	0.826	0.791	0.772	0.774	0.773	0.764	0.759	0.753	0.768	0.770	0.769	0.765	0.764	0.760	0.757	0.754	0.754	0.756
6	0.863	0.813	0.783	0.764	0.766	0.762	0.761	0.754	0.751	0.768	0.769	0.773	0.768	0.765	0.759	0.755	0.755	0.755	0.758
5	0.865	0.803	0.784	0.757	0.755	0.754	0.756	0.750	0.748	0.767	0.769	0.770	0.766	0.763	0.758	0.756	0.754	0.755	0.757
4	0.835	0.786	0.768	0.749	0.748	0.748	0.749	0.747	0.746	0.762	0.767	0.766	0.761	0.758	0.754	0.752	0.750	0.751	0.753
3	0.840	0.780	0.764	0.745	0.747	0.748	0.750	0.748	0.746	0.762	0.765	0.765	0.761	0.757	0.755	0.752	0.748	0.748	0.751
2	0.813	0.766	0.747	0.732	0.729	0.734	0.735	0.733	0.728	0.735	0.741	0.741	0.740	0.738	0.739	0.738	0.735	0.734	0.736
1	0.769	0.788	0.770	0.745	0.745	0.746	0.746	0.743	0.737	0.745	0.756	0.757	0.758	0.756	0.756	0.755	0.751	0.748	0.752

Coefficient-history pairs

Shared control

PC count K	Exp
8	0.730
7	0.732
6	0.735
5	0.736
4	0.736
3	0.734
2	0.728
1	0.731

Coefficient-history pairs

Common | A: matched PCA

PC count K	52	78	104	130	156	182	208	234	260	286	312	338	364	390	416	442	468	494	520
8	0.782	0.764	0.756	0.735	0.743	0.736	0.741	0.733	0.731	0.744	0.755	0.761	0.754	0.747	0.744	0.739	0.736	0.733	0.733
7	0.780	0.761	0.757	0.736	0.743	0.736	0.739	0.736	0.732	0.742	0.751	0.755	0.754	0.751	0.742	0.737	0.735	0.732	0.733
6	0.773	0.761	0.755	0.738	0.744	0.730	0.736	0.732	0.730	0.743	0.748	0.748	0.747	0.743	0.739	0.738	0.735	0.732	0.734
5	0.768	0.763	0.759	0.736	0.733	0.728	0.732	0.728	0.724	0.739	0.748	0.747	0.749	0.739	0.742	0.742	0.736	0.736	0.738
4	0.780	0.763	0.759	0.733	0.729	0.727	0.728	0.726	0.719	0.735	0.738	0.745	0.743	0.741	0.737	0.735	0.736	0.735	0.737
3	0.776	0.763	0.757	0.734	0.727	0.732	0.738	0.732	0.725	0.733	0.740	0.741	0.741	0.739	0.734	0.730	0.728	0.727	0.729
2	0.780	0.766	0.752	0.730	0.731	0.739	0.738	0.734	0.727	0.731	0.738	0.740	0.740	0.738	0.734	0.730	0.727	0.726	0.728
1	0.767	0.781	0.770	0.739	0.740	0.737	0.735	0.729	0.721	0.729	0.738	0.736	0.736	0.735	0.735	0.733	0.731	0.727	0.728

Coefficient-history pairs

Common | B: expanding PCA

PC count K	52	78	104	130	156	182	208	234	260	286	312	338	364	390	416	442	468	494	520
8	0.816	0.788	0.772	0.758	0.765	0.758	0.756	0.751	0.742	0.754	0.759	0.759	0.756	0.756	0.750	0.744	0.741	0.737	0.739
7	0.818	0.786	0.774	0.758	0.763	0.762	0.758	0.752	0.746	0.758	0.765	0.766	0.764	0.763	0.757	0.751	0.748	0.745	0.746
6	0.783	0.772	0.763	0.747	0.753	0.748	0.750	0.744	0.741	0.756	0.763	0.767	0.765	0.762	0.754	0.748	0.746	0.743	0.745
5	0.784	0.765	0.762	0.743	0.746	0.739	0.742	0.738	0.738	0.753	0.761	0.764	0.763	0.761	0.753	0.748	0.746	0.743	0.745
4	0.779	0.762	0.758	0.741	0.740	0.736	0.737	0.735	0.734	0.747	0.756	0.756	0.755	0.751	0.743	0.738	0.736	0.733	0.735
3	0.785	0.757	0.752	0.737	0.740	0.738	0.739	0.737	0.735	0.746	0.754	0.754	0.753	0.749	0.744	0.739	0.734	0.731	0.733
2	0.777	0.754	0.748	0.730	0.729	0.731	0.732	0.728	0.724	0.730	0.739	0.737	0.737	0.734	0.732	0.729	0.726	0.724	0.726
1	0.766	0.778	0.767	0.739	0.740	0.737	0.735	0.730	0.722	0.729	0.738	0.736	0.736	0.735	0.735	0.733	0.731	0.727	0.729

Coefficient-history pairs

Shared control

PC count K	Exp
8	0.717
7	0.720
6	0.719
5	0.720
4	0.720
3	0.721
2	0.718
1	0.720

Coefficient-history pairs

Partial | A: matched PCA

PC count K	52	78	104	130	156	182	208	234	260	286	312	338	364	390	416	442	468	494	520
8	0.797	0.785	0.761	0.741	0.752	0.749	0.748	0.737	0.731	0.742	0.749	0.753	0.745	0.740	0.739	0.733	0.730	0.727	0.728
7	0.787	0.784	0.764	0.740	0.744	0.743	0.747	0.739	0.729	0.742	0.747	0.752	0.748	0.744	0.739	0.732	0.730	0.730	0.730
6	0.776	0.784	0.759	0.737	0.743	0.742	0.744	0.732	0.730	0.744	0.744	0.745	0.740	0.736	0.736	0.732	0.730	0.728	0.729
5	0.772	0.786	0.758	0.733	0.733	0.738	0.742	0.732	0.729	0.745	0.743	0.742	0.741	0.733	0.740	0.737	0.735	0.735	0.737
4	0.782	0.776	0.754	0.733	0.729	0.737	0.732	0.726	0.718	0.734	0.735	0.742	0.737	0.735	0.734	0.732	0.733	0.733	0.736
3	0.779	0.766	0.756	0.733	0.728	0.739	0.742	0.735	0.727	0.734	0.739	0.737	0.737	0.734	0.732	0.730	0.728	0.728	0.730
2	0.793	0.773	0.755	0.733	0.732	0.743	0.741	0.735	0.726	0.731	0.737	0.737	0.737	0.735	0.736	0.730	0.729	0.729	0.730
1	0.768	0.787	0.772	0.747	0.743	0.741	0.740	0.733	0.724	0.732	0.740	0.739	0.739	0.738	0.738	0.735	0.733	0.731	0.732

Coefficient-history pairs

Partial | B: expanding PCA

PC count K	52	78	104	130	156	182	208	234	260	286	312	338	364	390	416	442	468	494	520
8	0.815	0.802	0.773	0.765	0.765	0.762	0.763	0.747	0.734	0.745	0.748	0.750	0.747	0.748	0.744	0.740	0.740	0.737	0.740
7	0.830	0.796	0.773	0.759	0.764	0.763	0.756	0.750	0.740	0.750	0.758	0.759	0.754	0.757	0.753	0.748	0.747	0.745	0.747
6	0.798	0.782	0.765	0.748	0.753	0.751	0.754	0.744	0.737	0.751	0.757	0.761	0.757	0.757	0.752	0.747	0.745	0.743	0.745
5	0.799	0.773	0.766	0.747	0.749	0.745	0.747	0.743	0.737	0.748	0.755	0.758	0.757	0.755	0.752	0.749	0.747	0.746	0.747
4	0.792	0.767	0.764	0.747	0.743	0.744	0.740	0.739	0.734	0.746	0.752	0.751	0.750	0.749	0.743	0.739	0.738	0.736	0.738
3	0.797	0.766	0.759	0.742	0.740	0.744	0.742	0.737	0.733	0.744	0.751	0.750	0.750	0.746	0.742	0.738	0.736	0.734	0.736
2	0.788	0.759	0.749	0.731	0.728	0.736	0.736	0.730	0.725	0.729	0.735	0.733	0.734	0.731	0.732	0.730	0.727	0.726	0.728
1	0.767	0.785	0.770	0.745	0.744	0.741	0.740	0.733	0.725	0.732	0.741	0.739	0.740	0.738	0.737	0.736	0.733	0.731	0.732

Coefficient-history pairs

Shared control

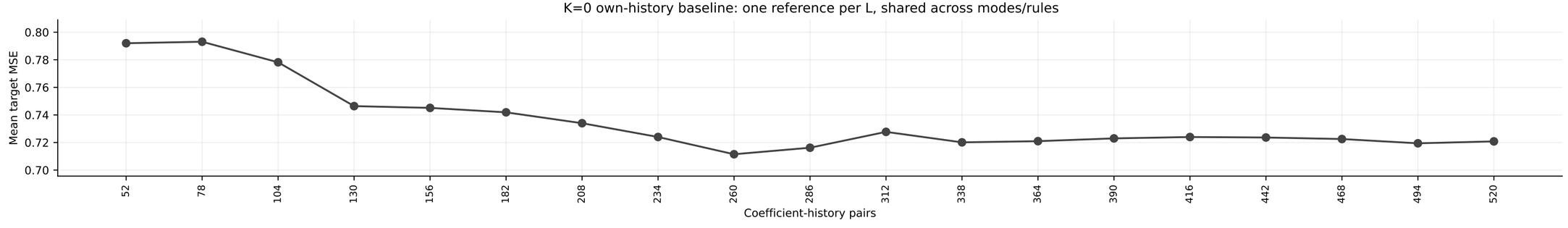
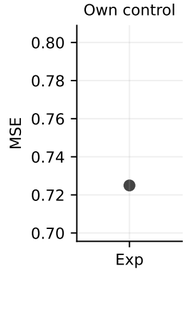
PC count K	Exp
8	0.719
7	0.722
6	0.723
5	0.725
4	0.724
3	0.724
2	0.720
1	0.723

Coefficient-history pairs

Mean target MSE (squared log units)

0.0 0.2 0.4 0.6 0.8

K=0 own-history baseline: one reference per L, shared across modes/rules



Own control